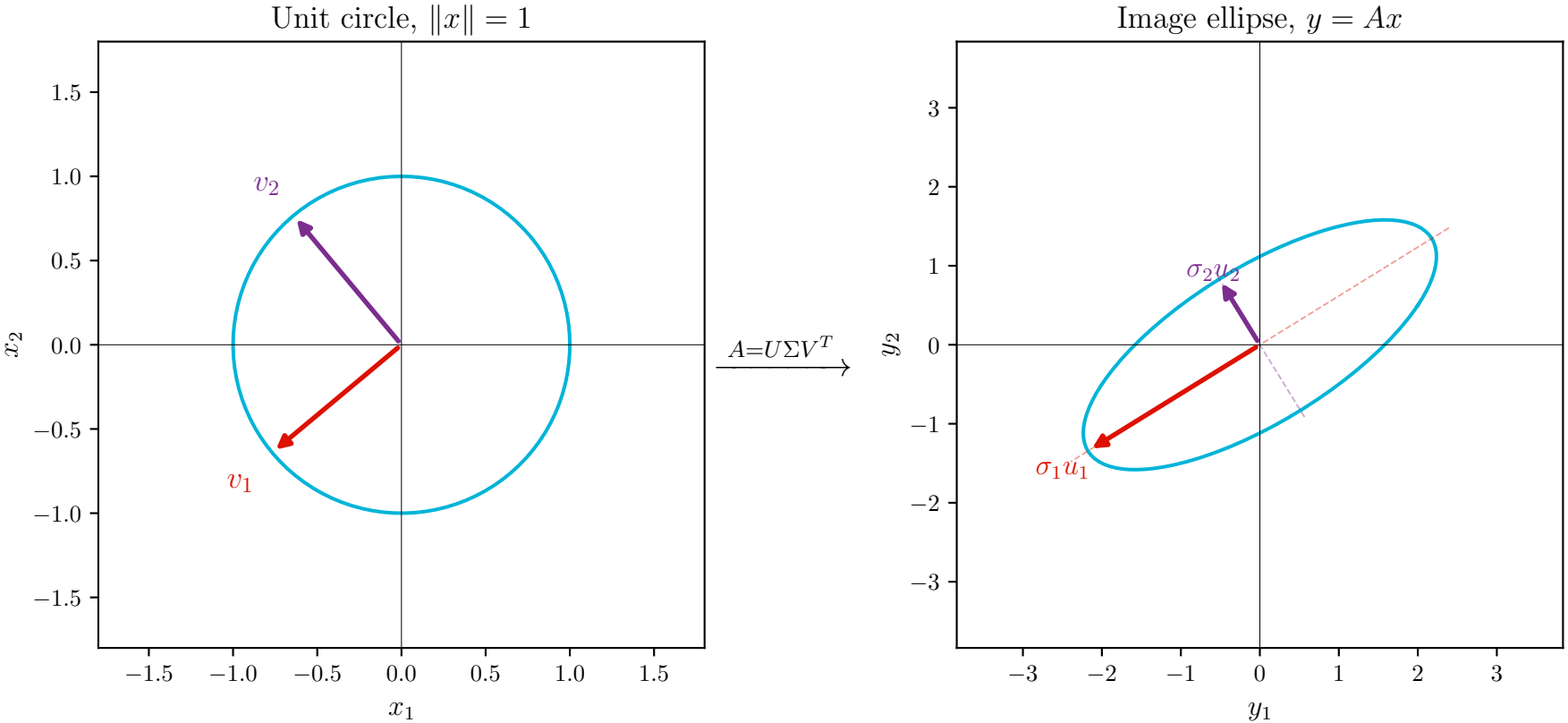
Unit circle, $\|x\| = 1$
v_2
v_1
x_1
x_2
$A = U\Sigma V^T$
Image ellipse, $y = Ax$
$\sigma_2 u_2$
$\sigma_1 u_1$
y_1
y_2